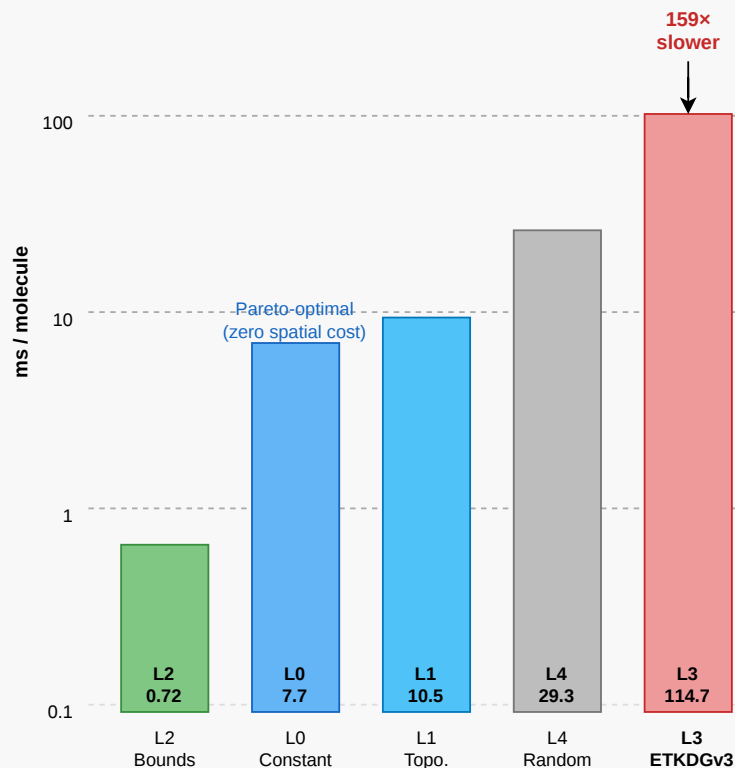


Computational Cost of Distance Representation Levels

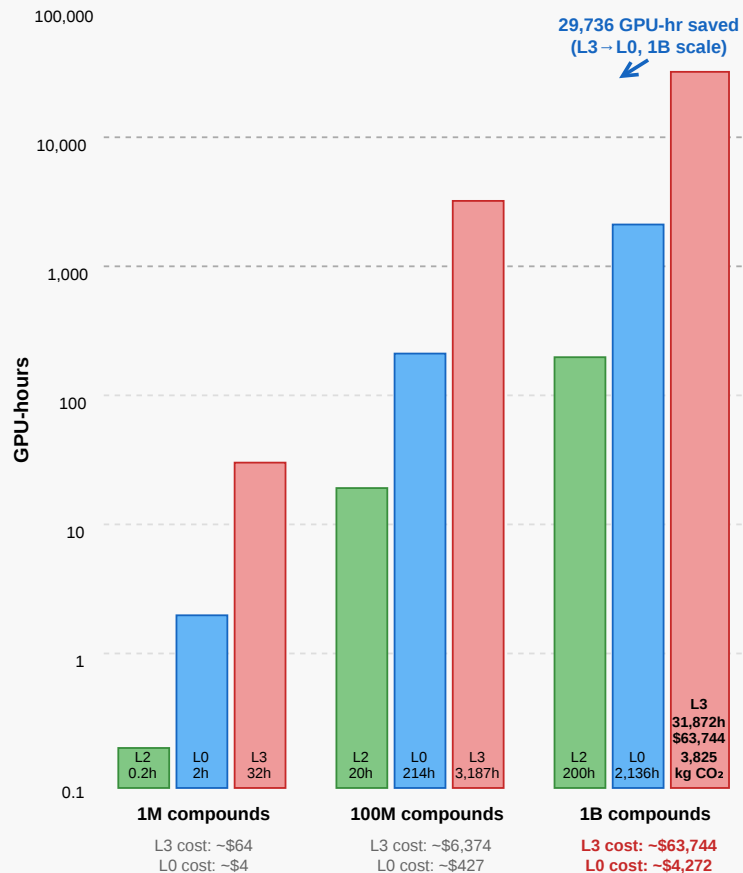
Per-molecule preprocessing time (ms/mol)

Log scale — ETKDGV3 is 159× slower than L2, and 15× slower than L0



Total GPU-hours at virtual screening scale

Log scale · Cost at \$2/GPU-hr · CO₂ at 0.4 kg/kWh with 300 W GPU



L2: Bounds (no conformer)

L3: ETKDGV3 (O(N³) conformer)

L0: Constant (zero spatial cost)

★ L3 → L0 at 1B scale: saves \$59,472 and 3,568 kg CO₂